Term Project - Wine Quality Data Analysis

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Introduction

The purpose of this project is to analyze the Wine Quality dataset, from UCI Machine Learning Repository, using both descriptive and predictive analytics. The goal is to identify the chemical features of red and white wine that influence wine quality and to build an interactive dashboard in Power BI that provide insights and enable drill-down exploration.

In addition, machine learning models were trained to predict wine quality, and their performance was evaluated using appropriate metrics.

This report summarizes the dataset, methods, preprocessing steps, descriptive statistics, predictive modeling, dashboard design, and key findings.

Description of the Dataset

The dataset used in this project is the Wine Quality dataset from the UCI Machine Learning Repository.

https://archive.ics.uci.edu/dataset/186/wine+quality

It contains measurements of physicochemical properties of red and white wines, along with a quality score from 0 to 10.

The dataset includes two separate files:

- winequality-red.csv with 1,599 samples (~25%),
- winequality-white.csv with 4,898 samples (~75).
- Total: 6,497 samples

Feature	Description	Type
Fixed Acidity	Tartaric acid concentration	Numeric
Volatile Acidity	Acetic acid concentration	Numeric
Citric Acid	Indicator of wine freshness	Numeric
Residual Sugar	Remaining sugar content after fermentation	Numeric
Chlorides	Salt concentration	Numeric
Free Sulfur Dioxide	Free SO_2 preservative level	Numeric
Total Sulfur Dioxide	Total amount of SO_2 preservative	Numeric
Density	Mass per unit volume	Numeric
pH	Measure of acidity	Numeric
Sulphates	Stabilizing agent affecting taste	Numeric
Alcohol	Alcohol percentage by volume	Numeric
Type	Wine category (red or white)	Categorical
Quality	Sensory score (0-10), used as target variable	Ordinal

Target Variables

- quality_label (binary classification)
  - 1 = High quality (>= 7)
  - 0 = Low / Average quality
- Quality (regression)

- Integer from 3 to 9

The dataset contains no missing values and is relatively clean but does contain outliers, especially in residual sugar, sulfur dioxide, and alcohol levels.

Dashboard Selection

Four pages were developed in Power BI to communicate insights clearly:

Page 1: Overview

A Summary including:

- Total Wines, average quality, high quality %, and average alcohol
- Distribution of red vs white wines
- Distribution of red vs white wines by quality and type
- Quality distribution by type
- Slicers for type and quality

Page 2: Descriptive / EDA

Investigates relationship between features:

- Correlation matrix
- Top 5 correlated features bar chart
- Boxplots of alcohol by quality
- Scatterplot of alcohol vs quality
- Slicer for type
- Insight annotations describing key findings

Page 3: Predictive / ML

Communicate model performance:

- Model comparison table (Accuracy, F1, R2, RMSE)
- Features by importance bar chart
- Confusion matrix
- Highlight top predictive feature
- Key Influencers visual

Page 4: Details

- Table with the full dataset

These pages collectively address descriptive and predictive insights.

Leveraging descriptive statistics

1. Numerical Feature Summary

Descriptive statistics were computed for all numeric features, including mean, median, mode, minimum, maximum, range, standard deviation, skewness, and kurtosis.

Key observations:

- Alcohol content: Red wines tend to have higher alcohol levels on average compared to white wines.
- Volatile acidity: Slightly positively skewed, meaning most wines have lower values, but a few wines have higher acidity.
- Residual sugar: Shows positive skew due to some very sweet wines.
- Quality: Approximately normally distributed, though white wines show slight left skew.

Visual summaries:

- Histograms of numeric features confirm skewness patterns.
- Boxplots show spread and highlight outliers.
- Example: alcohol and citric acid have relatively clean distributions, while residual sugar and sulfur dioxide measures have more extreme values.

2. Categorical Feature Summary

- The type of column has two categories:
  - Red: 1599 samples
  - White: 4898 samples
- White wines dominate the dataset (approximately 75% of total).
- Bar plot visualization of wine types provides a quick visual overview of this imbalance.

3. Correlations

- Pearson correlation was computed for all numeric features.
- Top correlations with quality:
  - Alcohol → positive correlation: Higher alcohol content generally relates to higher quality.
  - Volatile acidity → negative correlation: Higher acidity is usually associated with lower quality.
  - Sulphates → positive correlation: Moderate positive effect on quality.

Visual summary:

- Correlation heatmap highlights relationships among all numeric variables.
- Patterns observed in the heatmap confirm numerical correlation analysis and provide a visual guide for feature relationships.

4. Outliers

- Detected using boxplots and IQR method.
- Features with notable outliers: residual sugar, free sulfur dioxide, and total sulfur dioxide.

- Alcohol and citric acid distributions are relatively clean with fewer extreme points.
- Outliers should be considered in modeling or transformation if required.

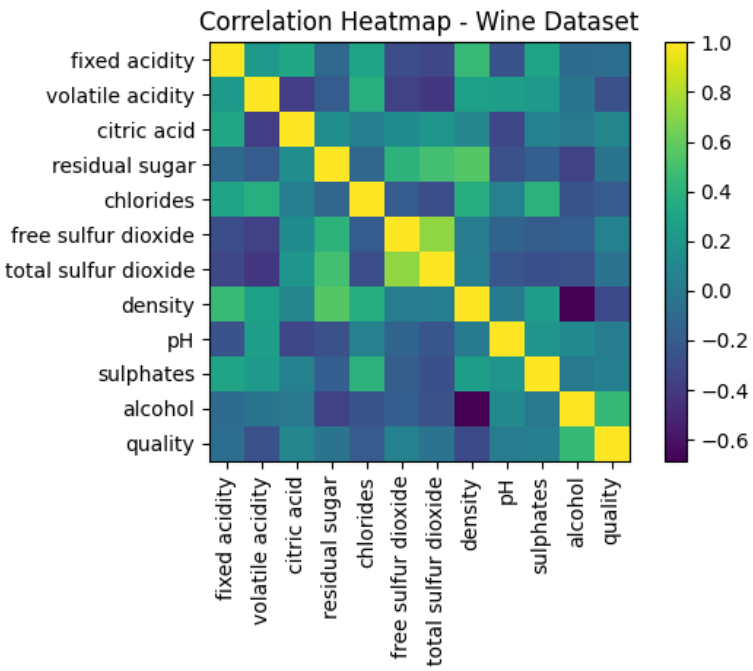
5. Visual Summary

- Histograms show the distribution of all numeric features.
- Boxplots reveal spread and potential outliers.
- Scatterplots and pair plots illustrate bivariate relationships between key features and wine quality.
- Observations from visualizations:
    - Volatile acidity vs quality confirms the negative correlation visually.
    - Alcohol content vs quality and wine type confirms higher alcohol in red wines.
    - Pair plots help identify feature interactions and clusters in the data.

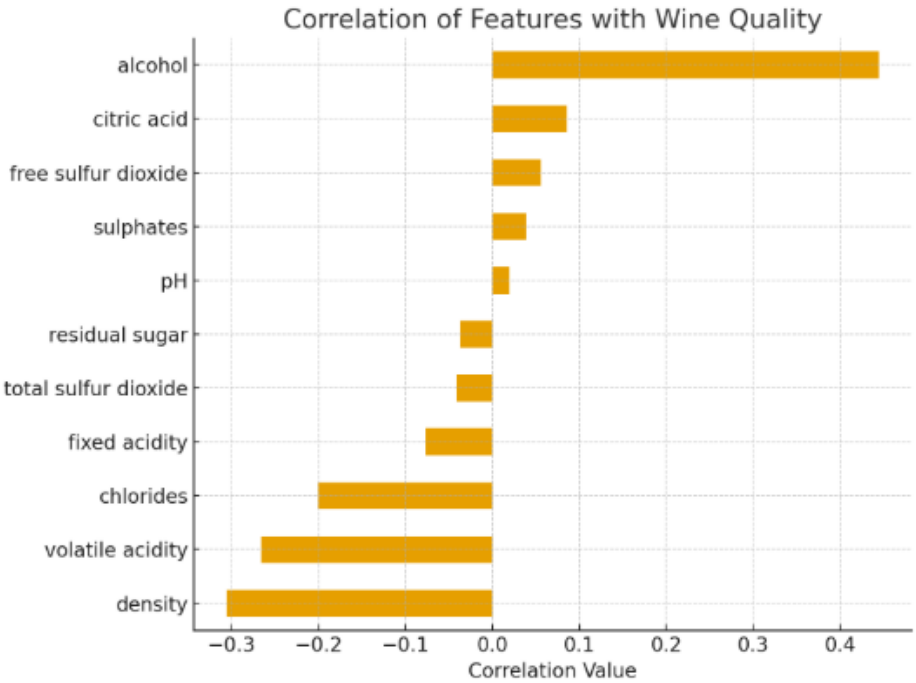
<u>Type of Graphs and Reasons:</u>

1. Correlation Heatmap

The first visualization is a correlation heatmap that displays pairwise correlation coefficients between the wine dataset's chemical properties and quality scores. This graph is used to provide a comprehensive overview of the linear relationships across all variables simultaneously. The color-coded scale allows for quick identification of strong positive or negative associations. In this context, the heatmap highlights that alcohol content exhibits the strongest positive correlation with wine quality, whereas density shows the strongest negative correlation. This visual summary supports variable selection and guides further exploratory analysis.

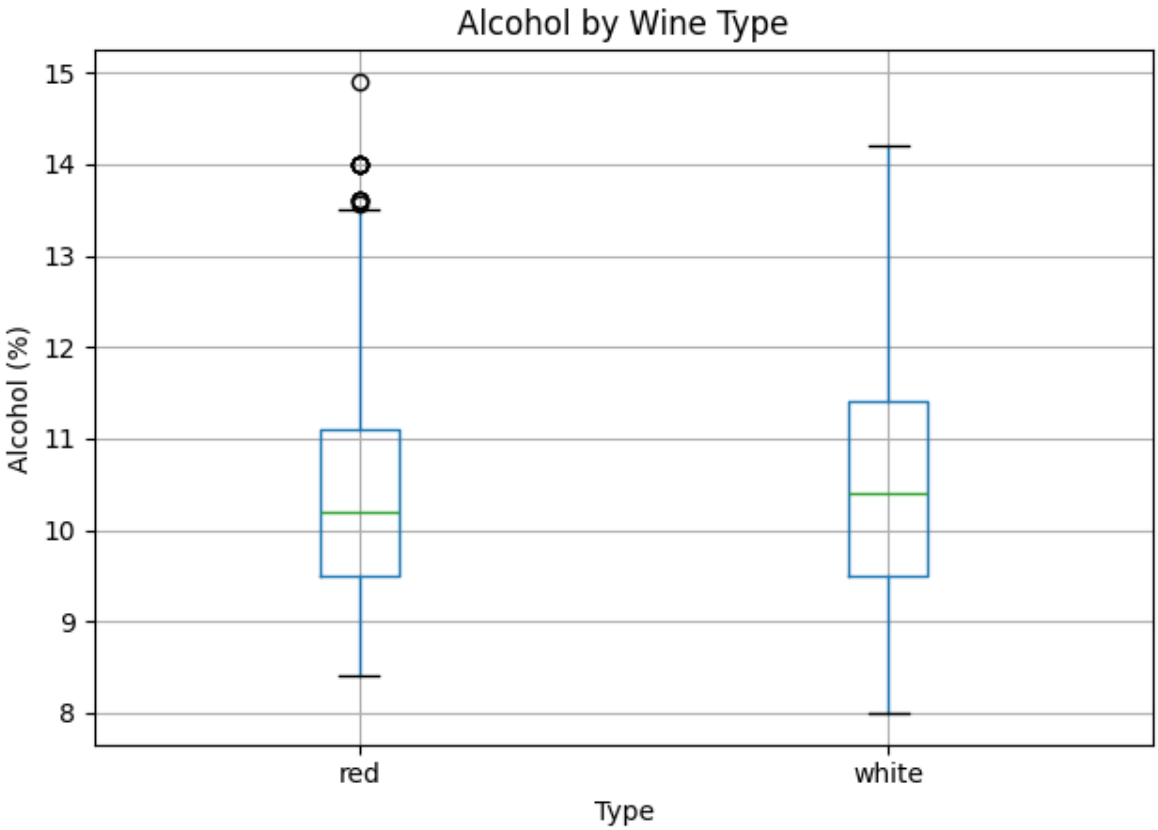


2. Correlation Bar Chart (Features vs. Quality)
The second graph is a correlation bar chart focusing specifically on the correlation values between each feature and wine quality. Unlike the full heatmap, this chart isolates the quality-related correlations, making it easier to compare the relative predictive strength of each variable. The bar chart format is appropriate because it clearly ranks features by their correlation magnitude, emphasizing that alcohol content is the most influential predictor among the variables.



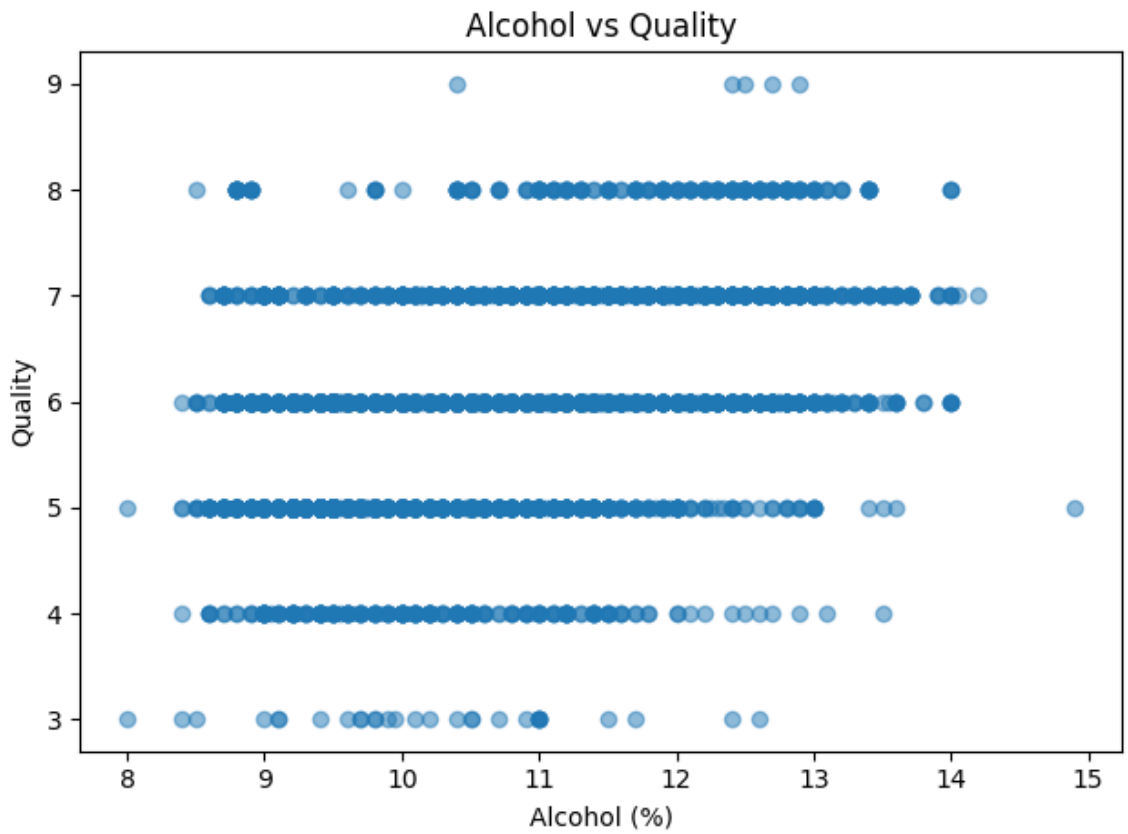
3. Box Plot of Alcohol Content by Quality Score
The third visualization is a box plot illustrating the distribution of alcohol levels across different wine quality categories. This chart is used to examine how alcohol content varies with quality and to identify patterns in the central tendency, spread, and presence of outliers. Box plots are well suited for comparing distributions across groups, and in this case, they reveal an upward trend in alcohol content for higher-quality wines. This supports the correlation results and provides distributional insight

beyond summary statistics.



4. Scatter Plot of Alcohol and Quality

The final graph is a scatter plot showing the relationship between alcohol content and wine quality at the individual data-point level. This visualization is used to assess the form, direction, and strength of the relationship between the two variables. The scatter plot demonstrates a general upward trend, indicating that higher alcohol levels tend to correspond with higher quality scores. It also provides visibility into variability within each quality level, complementing the box plot and confirming the positive relationship identified in the correlation analyses.



Page 3: Predictive / Machine Learning Analysis

The Predictive / ML page summarizes the performance of the machine learning models developed to predict wine quality and identifies the features that have the strongest influence on the target variable. Both **classification** and **regression** approaches were used, and the results were visualized in Power BI using tables, charts, and Key Influencer visuals to communicate insights clearly.

1. Model Performance Comparison

A comparison table was created to present the performance metrics of all models used. For the classification task (predicting *high-quality* vs *low/average* wine using quality_label), the primary metrics were Accuracy, Precision, Recall, and F1-Score. For the regression task (predicting the exact quality score), R^2, RMSE, and MAE were used.

The model comparison allowed us to quickly identify the strongest model for each prediction type:

- **Classification:** The Logistic Regression and Random Forest models performed well, with Random Forest producing the highest accuracy and F1-score due to its ability to model nonlinear interactions.

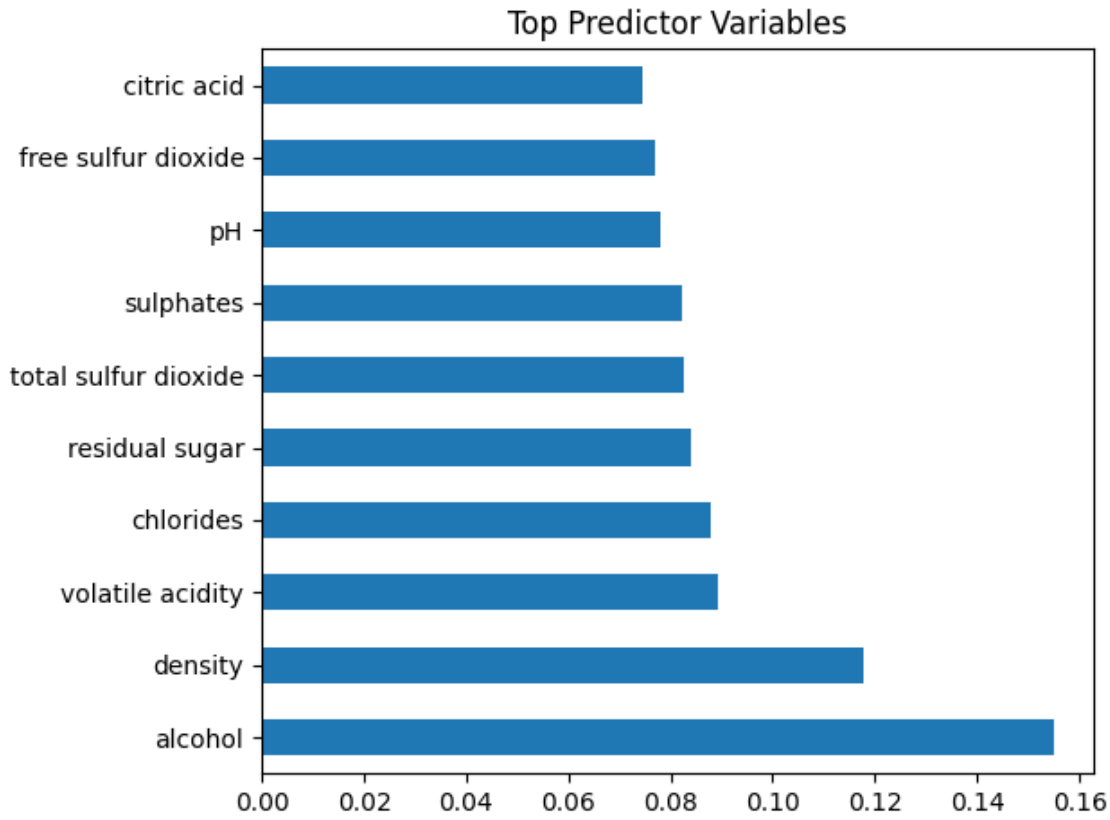
- **Regression:** Random Forest Regressor achieved the highest R^2 and lowest RMSE, indicating superior predictive power compared to Linear Regression.

The comparison table provides an at-a-glance summary of these results, helping users interpret model performance without needing to review raw metrics elsewhere.

2. Feature Importance Visualization

A bar chart was created to display the **feature importance scores** from the tree-based models (Random Forest).
This visualization shows which chemical properties contribute the most to predicting wine quality.

Key findings from the feature importance chart include:

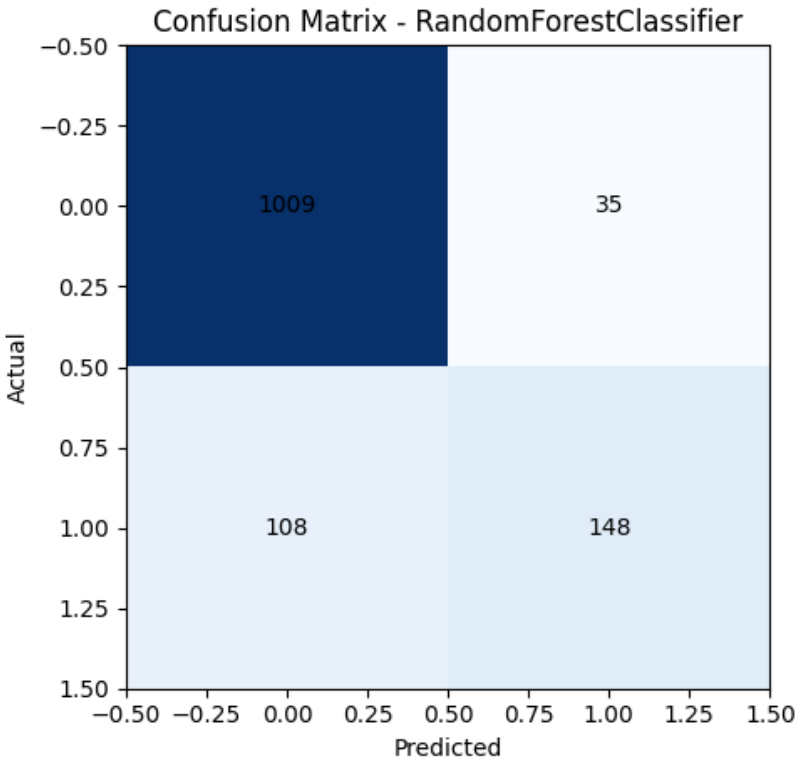


- **Alcohol** is the strongest predictor of wine quality across models.
- **Volatile acidity** and **sulphates** also play major roles, aligning with correlations seen in the EDA results.
- **Density** and **total sulfur dioxide** contribute moderately, while features like chlorides and residual sugar have smaller but still noticeable influence.

This visual connects the ML results with earlier descriptive insights, reinforcing the consistency of alcohol and acidity effects on quality.

3. Confusion Matrix

A confusion matrix was displayed for the final classification model to show how accurately the model distinguishes high-quality wines from lower-quality ones.



The matrix provides:

- **True Positives (correctly predicted high-quality wines)**
- **True Negatives (correctly predicted low/average wines)**
- **False Positives and False Negatives**

The confusion matrix helps interpret *how* the model performs—not just its overall accuracy.
For example, the model predicts low-quality wines very reliably but shows slightly more difficulty identifying rare high-quality samples, which is expected due to class imbalance (only ~20% of wines are labeled high-quality).

4. Highlighting the Top Predictive Feature

Power BI visuals were used to highlight the **single most impactful predictor**, which is **Alcohol**.
This matches:

- The strongest positive correlation in the heatmap
- The most important feature in Random Forest

- The clearest visual trend across scatterplots and boxplots

The consistency across multiple techniques strengthens the conclusion that **alcohol content is the primary driver of wine quality** in this dataset.

5. Key Influencers Visual

The **Key Influencers** visualization was included to provide an interpretable, high-level summary of:

- Which variables increase the likelihood of a wine being high quality
- How changes in each feature impact predicted quality
- How sub-segments (clusters) behave differently

Insights from this visual include:

- Higher alcohol levels significantly increase the probability of a wine being high quality.
- Higher volatile acidity sharply decreases quality predictions.
- For specific subgroups (e.g., white wine with elevated sulphates), the impact of certain features becomes stronger or weaker.

This visual bridges ML outputs and business/industry interpretability by explaining *why* the model makes certain predictions.

Summary

Overall, the Predictive / ML page provides a comprehensive and accessible overview of the wine quality prediction models.
The combination of performance tables, feature importance charts, confusion matrix, and Key Influencers visual creates a clear narrative:

Alcohol is the strongest predictor of wine quality, while acidity-related metrics (volatile acidity, sulphates, density) also play critical roles, and the Random Forest model delivers the most accurate predictions.

Conclusion

This project provided a clear understanding of the factors that influence wine quality by combining descriptive statistics, visual exploration, and machine learning. The analysis showed that alcohol, volatile acidity, and sulphates are the most significant predictors of quality, with alcohol having the strongest positive impact. Predictive models were also built, and the Random Forest model delivered the best performance for both classification and regression tasks. Overall, the project demonstrates how data analytics and predictive modeling can effectively support quality assessment and decision-making in real-world datasets.